

Question	Answer	Marks
8(a)	electric and magnetic fields at right-angles to one another (<i>may be shown on a clearly labelled diagram</i>)	B1
	particle enters fields (with velocity) normal to the (two) fields (<i>may be shown on a clearly labelled diagram</i>)	B1
	no deviation for particles with selected velocity	B1
8(b)	magnetic force equals/is the centripetal force	C1
	$Bqv = mv^2/r$	C1
	$M = Bqr/v$ $= (94 \times 10^{-3} \times 1.6 \times 10^{-19} \times 0.075) / (3.4 \times 10^4)$	M1
	<u>division</u> by 1.66×10^{-27} shown, to give $m = 20 \text{ u}$	A1
8(c)	sketch: semicircle clear (in same direction)	B1
	with larger radius	B1

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9(c)	(i) 1.5 g	1